

Foundations of Cybersecurity

Module 1

Welcome to the exciting world of cybersecurity

Get started with the certificate program

Welcome to the Google Cybersecurity Certificate

Google Cybersecurity Certificate overview

Course 1 overview

Welcome to module 1

Start the path to cybersecurity

Highly/inspiring and tips

Google Cybersecurity Certificate participant entry survey

Introduction to cybersecurity

Start the path to cybersecurity

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Responsibilities of an entry-level cybersecurity analyst

What's a day in the life of a security engineer analyst

Common cybersecurity terminology

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Core skills for cybersecurity professionals

Core skills for cybersecurity professionals

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Module 2

The evolution of cybersecurity

Module 3

Protect against threats, risks, and vulnerabilities

Module 4

Cybersecurity tools and programming languages

Course 1 overview

WELCOME

to Course 1

Hello, and welcome to **Foundations of Cybersecurity**, the first course in the Google Cybersecurity Certificate. You've begun an exciting journey!

In this course, you will learn the primary job responsibilities and core skills of those who work in the field of cybersecurity. You will explore the eight Certified Information Systems Security Professional (CISSP) security domains, various security frameworks and controls, as well as a foundational security model called the confidentiality, integrity, and availability (CIA) triad. You will also be introduced to some common tools used by security analysts that help protect organizations and people alike.

Certificate program progress

The Google Cybersecurity Certificate program has nine courses. **Foundations of Cybersecurity** is the first course.

Foundations of Cybersecurity

Protect against threats, risks, and vulnerabilities

Cybersecurity tools and programming languages

Automate cybersecurity tasks with Python

Put it to work: Prepare for cybersecurity jobs

Advanced threat job search with AI

Secure the cloud

Secure the network

Secure the system

Secure the people

1. Foundations of Cybersecurity

2. Play it Safe: Manage Security Risks

3. Connect and Protect: Networks and Network Security

4. Tools of the Trade: Linux and SQL

5. Assets, Threats, and Vulnerabilities

6. Sound the Alarm: Detection and Response

7. Automate Cybersecurity Tasks with Python

8. Put it to Work: Prepare for Cybersecurity Jobs

9. Accelerate Your Job Search with AI

Course 1 content

Each course of this certificate program is broken into modules. You can complete courses at your own pace, but the module breakdowns are designed to help you finish the entire Google Cybersecurity Certificate in about six months. What's to come? Here's a quick overview of the skills you'll learn in each module of this course.

Module 1: Welcome to the exciting world of cybersecurity

Begin your journey into cybersecurity! You'll explore the cybersecurity field, and learn about the job responsibilities of cybersecurity professionals.

Module 2: The evolution of cybersecurity

You will explore how cybersecurity threats have appeared and evolved alongside the adoption of computers. You will also understand how past and present cyber attacks have influenced the development of the security field. In addition, you'll get an overview of the eight security domains.

Module 3: Protect against threats, risks, and vulnerabilities

You will learn about security frameworks and controls, which are used to mitigate organizational risk. You'll cover principles of the CIA triad and various National Institute of Standards and Technology (NIST) frameworks. In addition, you'll explore security ethics.

Module 4: Cybersecurity tools and programming languages

You'll discover common tools used by cybersecurity analysts to identify and eliminate risk. You'll learn about security information and event management (SIEM) tools, network protocol analyzers, and programming languages such as Python and SQL.

What to expect

Each course offers many types of learning opportunities:

Videos

Readings

Connect with other learners

Self-review activities

Interactive plug-ins

In-video quizzes

Practice quizzes

Graded quizzes

Tips for success

It is strongly recommended that you go through the items in each lesson in the order they appear because new information and concepts build on previous knowledge.

Participate in all learning opportunities to gain as much knowledge and experience as possible.

If something is confusing, don't hesitate to replay a video, review a reading, or repeat a self-review activity.

Use the additional resources that are referenced in this course. They are designed to support your learning. You can find all of these resources in the **Resources** tab.

When you encounter useful links in this course, bookmark them so you can refer to the information later for study or review.

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Give me a summary

Give me practice questions

Give me real-life examples

Ask me anything

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https://www.coursera.org/learn/foundations-of-cybersecurity/supplement/QBRd5/course-1-overview

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